CHAPTER

31

Pulpal Reactions

Pulp, a specialized connective tissue, is very sensitive to external stimuli. All dental procedures, extent may vary, threaten the integrity of the pulp. Any stimulus over the exposed dentinal tubules affects the pulp. The reaction of pulp, most of the times, is physiologic; however, depending upon the intensity of the stimulus, pathological changes do occur in pulp.

It is established that the contact of a dental material with dentin can induce odontoblastic alterations and transient inflammatory reactions. Earlier, the acidic components from zinc phosphate and unreacted monomers from composite restorations were considered injurious to the pulp. It is established that leakage of bacteria and their byproducts from the oral environment along the gaps in the cavity–restoration interface are more toxic than the material itself.

It is mandatory to know the comparative toxicity of dental materials, and techniques so that necessary measures can be undertaken to reduce their toxic effect. The operator should have thorough knowledge of the materials and techniques so as to use them cautiously; subsequently, minimizing pulpal reactions.

TERMINOLOGY

The commonly used terms in the context of safety of materials are as follows:

i. Biocompatibility

The term 'biocompatible' is defined as being harmonious with life and not having toxic or injurious effects on biological functions. In general, biocompatibility is measured on the basis of localized cytotoxicity (such as pulp and mucosal response), systemic responses, allergenicity and carcinogenicity.

The dental materials should not be harmful to the pulp and soft tissues. It should not contain toxic substance that can be absorbed into the circulation to cause a systemic response. It should have no carcinogenic potential and also should be free of potentially sensitizing agents that could cause an allergic response.

ii. Biomaterial

Biomaterial is defined as any substance, other than a drug, that can be used for any period as a part of a system that treats, augments or replaces any tissue, organ or function of a body. Dental materials, used in humans for short or long periods, fall in the category of biomaterial. The host environment for dental biomaterials is especially complicated because of the presence of bacteria and debris in the oral cavity and the corrosive properties of saliva and other fluids.

iii. Hazardous

A material is considered hazardous; when placed in contact, it has the potential to create harmful reaction in the body tissues. Hazardous materials are generally identified through screening tests; the material is placed in direct contact with either cultured cells or animal tissues.

iv. Risk

Risk implies the probability that the material would cause sufficient damage. A material, which is hazardous but do not pose a significant risk to the body (e.g. zinc oxide eugenol), is hazardous but not risky. Risk may be reduced because of diffusion or dilution of the offending components, biological barriers, or a lack of sufficient time of contact between the offending components and susceptible tissues.

PULP–DENTIN ORGAN

The dentin and the pulp must be considered as one organ (the pulp–dentin complex) because of the intimate relationship between the cellular tissue within the dentin and the peripheral pulp tissue. Structure and response of dentin to injury are largely functions of the odontoblasts and other cells in the

pulp, but these cells are dependent on the dentin for their protection and their state of differentiation. Normal form and function of one cannot be maintained without the other. The embryonic dental papilla is responsible for the formation of this couple tissue. Hence, it is obvious that the response of the pulp to any restorative material will be influenced by its surrounding dentin also. The dentinal tubules occupy 20–39% of dentin, and the dentinal fluid within them represents about 22% of the total volume of dentin. Dentinal fluid in the tubules, which is continuous with the extracellular fluid of the pulp serves as a medium for relaying injurious agents to the pulp to induce an inflammatory response. Thus anything that contacts the living dentin can be carried into the pulp. Either positive hydrostatic pressure or negative osmotic pressure facilitates movement of fluid in the dentinal tubules; displacing odontoblastic processes and subsequently affecting the pulp.

Physiology

The pulp is a connective tissue, which is derived from the mesenchyme, the dental papilla. The pulp is characterized by its specialized odontoblasts on the periphery beneath the predentin. Beneath the odontoblastic layer is the *'cell-poor zone'* or the *'Weil's layer'* composed principally of young fibroblasts. The central portion of the pulp consists of many young fibroblasts and occasional lymphocytes. It is richly supplied by thin walled but wide capillaries. Bundles of myelinated nerve fibers accompany the blood vessels. Fibers predominantly present are collagenous and reticular types. Lying in close proximity to the blood vessels are many undifferentiated mesenchymal cells and histiocytes. These are believed to transform into secondary odontoblasts that form reparative dentin following the death of primary odontoblasts.

The interstitial fluid of the pulp and the dentinal tubules are continuous and extend from dentino-enamel and dentino-cementum junction to the soft connective tissue of the pulp.

The pulp is vascularized as other organs of the body. A large arteriole passes through the root pulp to supply the coronal pulp. They branch as capillaries in the coronal subodontoblastic region. The blood capillaries have discontinuity in the endothelial walls. Such types of arrangements facilitate easy exchange of nutrients and the waste fluid. This exchange is important in case of pulp injury. The fluid flow from the pulp to exposed dentin is dependent on the hydraulic conductance of the dentinal fluid. Any reduction in this conductivity will reduce the dentin sensitivity.

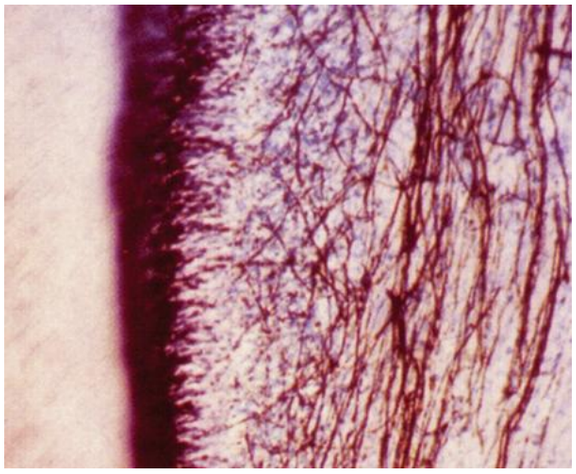


Fig. 31.1: Pulpal innervation

Myelinated and unmyelinated nerves enter the pulp through apical foramen (Fig. 31.1). Usually the myelinated fibers and unmyelinated Aδ-fibers carry pain impulses. The sensory and the sympathetic nerve endings are activated at an early stage of the inflammatory process and are the initiators of vasodilatation. This is the start of the protective response to the injury by increasing blood volume and vascular permeability in the affected areas. Both the sympathetic and sensory nerve fibers have effect on the pulpal circulation. The number of nerve fibers decreases with age, which explains the reduced sensitivity in older adults.

Dentin has an average of 45,000–90,000 tubules/mm^2 near the pulp, 30,000–35,000/mm^2 in the middle and 10,000–25,000/mm^2 at the periphery. At the pulp surface, the diameter of the tubules ranges from 2.5–3.0 μm and at the periphery, the diameter may be less than 1.0 μm. Each tubule is about 1.0–3.0 μm in diameter and 2.0–3.0 mm in length. The tubules have highly mineralized lining, the peritubular dentin, which is not found in the pulpal part of the dentin of the newly erupted teeth. The highly mineralized peritubular dentin should be distinguished from the other mineralized components of dentin, the intertubular dentin that has much less collagen than the peritubular dentin. The growth of peritubular dentin, may be age related, or due to restorative procedures, leads to obliteration of the tubules. The occluded dentinal tubules, referred to as dentin sclerosis, react differently to acid etching. (Etching time is modified to provide an adequate hybrid layer of collagen and resin.)

The interface dentin with irregular, often atubular dentin, forms a barrier between the physiologic secondary and tertiary dentin. This barrier reduces the permeability of the affected dentin and may make

it impermeable because the tubules from primary dentin do not cross the interface dentin. This phenomenon protects the pulp.

Branching of dentinal tubules differs as the number of tubules, depending upon the location within the tooth. In coronal dentin, the characteristic branching is found in peripheral 250 µm, where Y-shaped branches are evident. The fine branches are predominantly present in root dentin, but may be evident in other parts of the dentin such as cervical dentin.

Generally, dentin directly under the dentino-enamel junction is only 1.0% permeable, but permeability increases to 7.6% halfway to the pulp and up to 22.0% at the pulp surface. The dentinal tubules contain odontoblastic processes, which are direct extensions of the cell body of the odontoblasts underlying the predentin. It is these dentinal tubules and their contents that make operative dentistry so intimately related to the vitality of the pulp. The pulpal circulation maintains an intercellcular hydraulic pressure of about 24 mm Hg, which causes the fluid flow in the tubules, directed outwards from the pulp to the dentinoenamel junction. Chemical and bacterial products when introduced into unprotected dentinal tubules diffuse against the pressure gradient towards the pulp.

PULPAL RESPONSE TO RESTORATIVE MATERIALS

It is established that pulp reactions to the same irritant may vary not only between different persons but also between different teeth and between opposing teeth of the similar anatomic form.

The factors influencing pulpal responses are as follows:

a. Dentin Permeability

A variety of features affect dentin permeability; the main are:

i. ***Location:*** Dentin permeability varies in different areas of the same tooth; for example, it increases towards the pulpal side. This is because both the tubule diameter and the number increase towards the pulp chamber.

ii. ***Tubule diameter (diffusional area):*** The tubule diameter directly influences the wetness; subsequently, the hydrolytic dissolution of the restorative material.

iii. ***Smear layer:*** The presence or absence of the 1.0–5.0 mm thick layer of microcrystalline debris on the cut dentin surface (smear layer) controls dentin permeability. Presence of smear layer reduces permeability.

iv. ***Intratubular calcifications:*** Intratubular calcifications, viz. mineral deposits, collagen fibrils, proteins, etc. may reduce permeability.

v. ***Concentration and solubility of diffusing solutes:*** The substances with high molecular weight and size are less penetrating than the substances with low molecular size and weight. Solutes that are water soluble would show a rapid rate of penetration.

vi. ***Patency of the dentinal tubules:*** Sclerotic dentin is less permeable than the physiologic tubular dentin.

vii. ***Reparative dentin:*** Reparative dentin reduces the penetration of the solute.

viii. ***Remaining dentin thickness:*** The remaining dentin thickness of 2.0 mm provides an adequate insulating barrier against almost all the techniques and restorative materials. It has been shown that a 0.5 mm thickness of dentin reduces the toxicity level of a material to 75% and a 1.0 mm thickness may reduce over 90%.

b. Dentin Conditioning

Pre-conditioning of dentin by acids (etchants) increases dentin permeability by removing smear layer and enlarging tubule orifices. The effect depends upon the concentration and the duration for which the etchant is applied.

c. Age of the Patient

It is hypothesized that the inflammatory response of the older persons is less extensive and the reparative dentin formation is also less as compared to the younger individuals. The older pulps are far less defensive in resolving a lesion and resisting infection. However, a few authors have observed that age might not directly affect the pulpal response.

INTENSITY OF PULPAL RESPONSE

The pulpal reactions, as depicted histopathologically, are as follows:

1. Cellular Displacement

It is characterized by movement of odontoblasts and leukocytes into the dentinal tubules. This occurs in the first few days and may persist over 30 days. It provides evidence of the acuteness of initial response after the leukocytes are degenerated and other characteristics of acute inflammation have resolved.

2. Initial Infiltration of Inflammatory Cells

The inflammatory cells are infiltrated into odonto-blastic layer, zone of Weil and cell-rich zone; may also

infiltrate into deeper tissues. These features are arbitrarily graded as 0 to 3; merely on the basis of the number of displaced/inflammatory cells present. When the response represents only one or two displaced cells, a graded value of one-half is given. Rarely, the intensity of the response is so great, which is to be categorized in grade 4.

3. Resolving Inflammatory Cells

Eosinophils are grouped with the chronic inflammatory cells because they appear concomitantly in pulp lesions. As the lesion resolves, the acute inflammatory cells diminish in number and are replaced by smaller and deeper tissues. With extended post-operative intervals, the chronic inflammatory cells gradually disappear in a resolving lesion with the formation of reparative dentin.

The response usually takes 1 to 6 days; however, may take 20 days in some cases.

4. Special Histopathologic Features

a. *Abscess formation:* Occasionally, beneath cavity preparation, certain abscess-like conditions (dense accumulations of leukocytes between the odontoblastic layer and the predentin), may develop, which usually resolve the tissue. These characteristics are not necessarily localized, but sometimes they can be quite focal.

b. *Foci of necrosis:* Cavity preparation and placement of toxic material lead to loss of cellular detail, collapse of vascular channels and a paucity of inflammatory cells. These lesions get infiltrated by inflammatory cells; usually resolve with granulation tissue replacement and sometime undergo abscess formation.

c. *Regeneration of odontoblasts:* In case the lesion is resolved, the inflammatory cells may disappear and leave behind an atrophic or degenerated odontoblastic layer [foci completely lacking in primary odontoblasts (Fig. 31.2)]. In some instances, only regenerated odontoblasts are found and a distinction needs to be made between these two types of layers.

d. *Reparative (tertiary) dentin formation:* The greater the degree of initial response due to irritation caused by cutting and placing of a restorative material, the greater the incidence of reparative dentin formation (Fig. 31.3).

Reparative dentin seldom develops earlier than 30 days.

e. *Eosinophilic staining:* The tissues involved are inundated by an amorphous material similar to that seen in capillaries. It is assumed that this is a plasma transudate, which escapes through mildly injured capillaries.

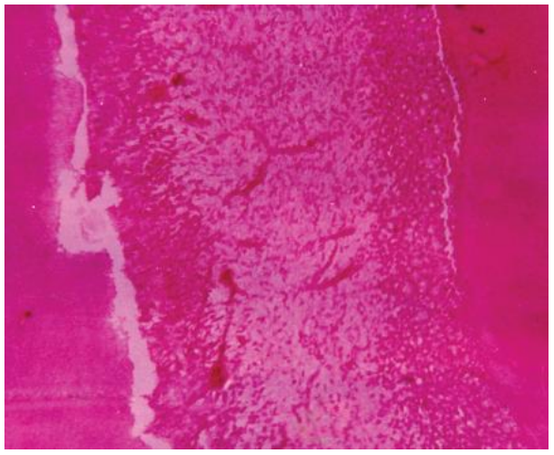


Fig. 31.2: Odontoblasts showing hydropic degeneration

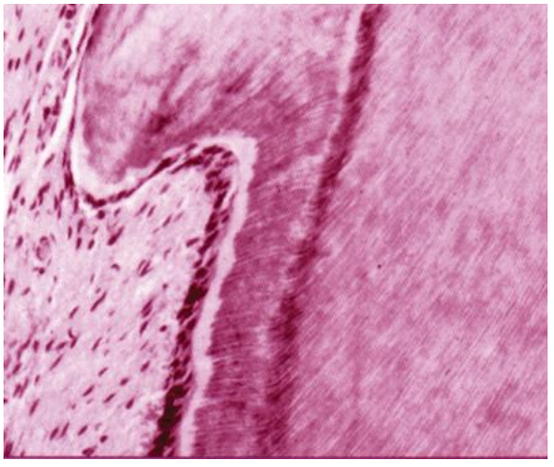


Fig. 31.3: Tertiary dentin formation

This characteristic is found either in operative techniques capable of producing a low order of irritation or on the periphery of many larger pulpal lesions away from the center of inflammation.

STAGES OF PULPAL INFLAMMATION

The reaction of the pulp to external stimuli is reflected in two broad processes: Inflammatory changes and secondary dentin formation.

Whenever the pulp is subjected to external stimuli, the immune system triggers inflammatory response to limit the tissue damage. The inflammation in pulp, the pulpitis, is similar to inflammation of any connective tissue in the body. The polymorphonuclear leukocytes are associated with acute reaction and also the mononuclear leukocytes, plasma cells and macrophages. Mast cells are only found in inflamed pulp, however, not in normal pulp. The inflammatory processes have been quantitatively divided into slight, moderate and severe.

i. *Slight reaction* is characterized by increased number of cells in the cell-free zone and in the adjacent pulpal tissue. These cells are similar to fibroblasts and undifferentiated cells. A few inflammatory cells are also observed. The increased number of capillaries, means blood flow is also increased. The irregularities in the odontoblastic layer are also observed (Fig. 31.4).

ii. *Moderate reaction* is characterized by more cells around the injury site. The mononuclear leukocytes and the neutrophils invade the odontoblast–predentin area. Some odontoblastic nuclei can be seen in dentinal tubules. The number of capillaries increased along with the blood flow. The occasional hemorrhage in odontoblastic or subodontoblastic zone is also observed (Fig. 31.5).

iii. *Severe reaction* is characterized by marked cellular infiltration, including abscess formation. The odontoblastic layer remains unidentified. This layer is either destroyed or greatly disrupted. The predentin is not formed. Numerous blood vessels are found in the tissues surrounding the cellular infiltration (Fig. 31.6).

Earlier, plus sign was used to indicate the degree of infiltration of inflammatory cells. The plus sign was also used to indicate the amount of secondary dentin formation.

Later, the inflammation and secondary dentin formation was graded as I^0–I^3 and D^{-1}–D^3, respectively.

- I^0: Absence of inflammation and no disturbance of the odontoblastic layer.
- I^1: Involvement of the odontoblastic layer only; including aspiration of the nuclear debris.
- I^2: Extension of inflammation to the subodontoblastic layer.
- I^3: Involvement of the central pulp.

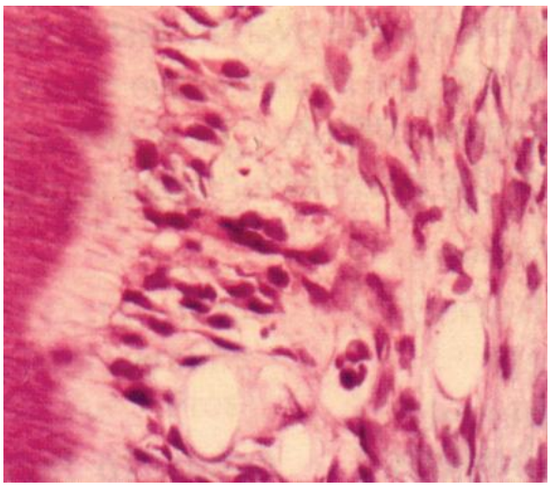


Fig. 31.4: Slight reaction: Odontoblastic layer with increase in number of capillaries

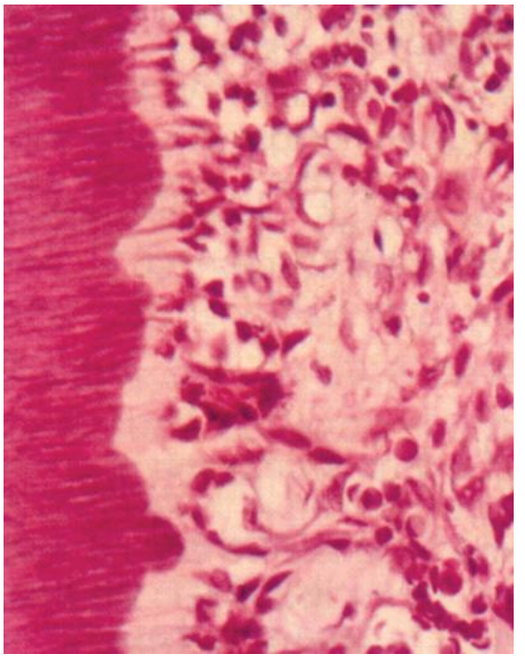


Fig. 31.5: Moderate reaction: Odontoblastic nuclei displaced

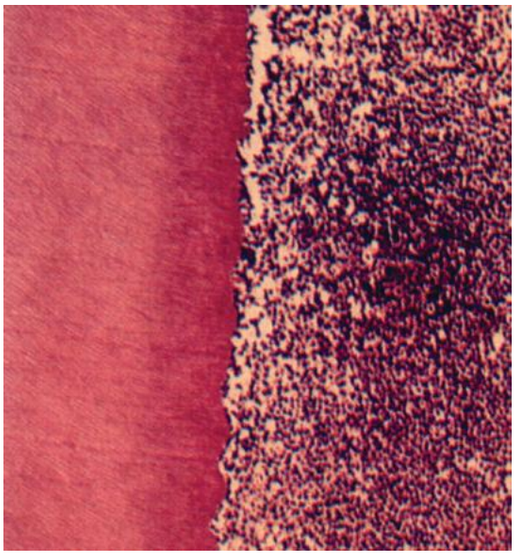


Fig. 31.6: Severe reaction: Mononuclear cells predominate

- D^{-1}: Absence of secondary dentin formation and destruction of odontoblasts.
- D^0: Normal appearance of odontoblastic layer with a normal width of predentin (20–30 μm).
- D^1: Slight amount of secondary dentin formation (35–60 μm)
- D^2: Moderate amount of secondary dentin formation (60–90 μm)
- D^3: Considerable amount of secondary dentin formation (>90 μm)

EVALUATION OF BIOCOMPATIBILITY

The aim of evaluating biocompatibility is to eliminate any product or a component of product that can damage the oral tissues. The term 'irritation' and 'sensitization' must be understood and differentiated. 'Irritation' is defined as an inflammation brought about without the intervention of an antibody or immune system; whereas, 'sensitization' is an inflammatory response requiring the participation of antibody system.

The restorative materials should pass all levels of biocompatibility testing before it is approved for final usage in the human oral cavity.

The tests are categorized as follows:

A. Primary Tests

a. *Cytotoxicity tests:* Cytotoxicity tests include the placement of a dental material directly onto the tissue culture cells. The cells react depending upon the components that leach from these materials. The toxic response can be quantified by measurement of biologic function, such as, DNA synthesis, protein synthesis, or the concentration of components that cause a decrease in cellular activity by 50% (TC 50 concentration). TC 50 concentration is inversely proportional to the potency of the cytotoxic response, i.e. lower the TC 50 concentration, higher is the potency.

b. *Genotoxicity tests:* Mammalian/non-mammalian cells, bacteria/yeasts/fungi are used to determine whether gene mutations, changes in chromosome structure, or other DNA changes are caused by the restorative materials.

B. Secondary Tests

a. *Systemic toxicity tests:* The test sample is administered daily to rats for 14 days, either by oral lavage or by dietary inclusion. If 50% of the animals survive, the product is considered safe.

b. *Inhalation toxicity tests:* These tests are performed on rats or guinea pigs in an exposure chamber. The spray material is released around the head of the animals for 30 seconds, each at 30 minutes intervals. After 10 consecutive exposures, the animals are observed for 4 days. If any animal dies within this period, the agent is considered toxic, otherwise safe.

c. *Skin irritation and sensitization tests:* The test material is held in contact with the shaved skin of albino rats for periods ranging from 24 (one exposure) to 90 days (with repeated exposures) for evaluating irritation effects.

d. *Implantation tests:* Various physical properties of the material, such as density, hardness and surface finish influence the character of the tissue response. The implant material is packed in plastic tubes and put into subcutaneous tissues or directly into bones in drilled holes. The reactions are evaluated.

C. Preclinical Usage Tests

a. *Pulp and dentin usage test:* The material is placed in contact with open dentin 1.0 mm from the pulp of rodents, and evaluated after a given period of time. The specimens are examined regarding inflammatory response, reparative dentin formation and the number of micro-organisms entrapped in the surrounding cavity walls and cut dentinal tubules. If an inflammatory response is produced, the time required for its disappearance is also measured.

b. *Pulp capping and pulpotomy usage tests:* The procedure is similar to the previous ones, except that the pulp is exposed. The animals are evaluated for dentin bridge formation, its quality and structure.

c. *Endodontic usage tests:* The pulp is completely removed and replaced by the obturating material. The degree of inflammation is evaluated in the periapical tissues.

PULPAL REACTION TO COMMONLY USED RESTORATIVE MATERIALS

A. Zinc Oxide Eugenol

Zinc oxide eugenol (ZOE) is frequently used material; considered bland, and even therapeutic to the pulp. Since zinc oxide eugenol has a low compressive strength, different additives are added to reinforce the original formulation. These are having better physical properties; however, more irritating to pulp.

Zinc oxide when mixed with eugenol leads to formation of zinc eugenolate matrix. The zinc eugenolate units are held together by van der Waal's forces and particle interlocking. When exposed to aqueous media, such as saliva or dentinal fluid, hydrolysis of zinc eugenolate occurs, yielding eugenol and zinc hydroxide. Eugenol liberated from zinc eugenolate can diffuse through dentin and into the saliva. The concentration of 10^2 mol/L of eugenol can be observed below the fillings in dentin and 10^4 mol/L in the pulp; remain maintained for more than one week. Calcium in the dentinal tubules chelates eugenol, limiting its ability to diffuse through dentin. Eugenol also binds with the organic matrix of dentin, especially collagen, which slows the diffusion rate. Modified ZOE cements have demonstrated less release of eugenol and fewer cytotoxic effects. It has been observed that acid-etched dentin may facilitate

diffusion of potentially toxic amount of eugenol to the pulp.

Eugenol is bactericidal at relatively high concentration of 10^2 to 10^3 mol/L. Even lower concentrations can inhibit cell respiration and cell division. Inhibitory eugenol concentrations are significantly higher than the one required for anti- inflammatory effects. The zinc oxide eugenol placed in direct contact with the pulp tissue results in chronic inflammation and necrosis. However, when placed against dentin, the cytotoxic effects are nil.

Various mechanisms that explain the cytotoxicity of eugenol are:

- Eugenol can be oxidized by peroxidase enzymes, the product formed is toxic to hepatocytes.
- Possess high affinity to plasma membranes because of its lipid solubility, which can cause cell damage.
- Eugenol has shown to uncouple oxidative phosphorylation in mitochondria.
- Protects tissue from damage by inhibiting neutrophil function and chemotaxis.
- Inhibits prostaglandin and leukotriene synthesis, which are important mediators of inflammation by increasing blood flow and vascular permeability and lowering pain threshold (At concentration levels of 10^4 mol/L, eugenol has shown to inhibit prostaglandin synthesis and sensory nerve excitability. At low concentrations, the intradental nerve activity is blocked reversibly, just like a local anesthetic; whereas, at high concentrations, nerve conduction is irreversibly blocked, indicating a neurotoxic effect).
- Eugenol causes vasodilation and decreases the response of these vessels to norepinephrine and histamine. Vasodilation would result in prevention of toxic accumulations and rapid removal of irritants.

The histologic response of a normal pulp to zinc oxide eugenol placed in standard cavity depths is a mild chronic inflammatory cell infiltration. At five interval weeks, there is a persistence of inflammatory cells underlying the cavity floor with a little reparative dentin formation. The effect of eugenol is dose dependent.

Effects of eugenol	
Toxic (high dose)	*Beneficial (low dose)*
• Induces cell death	• Inhibits white cell chemotaxis
• Unknown vascular effect	• Inhibits prostaglandins synthesis
• Inhibits cell growth and respiration	• Inhibits nerve activity

B. Zinc Phosphate Cement

Zinc phosphate cement when used as a base, has a low toxicity level; but quite toxic when used as a luting agent. Zinc phosphate cement is irritating because of its low pH and the rapid penetration of its lower molecular weight phosphoric acid into the dentinal tubules. The hydraulic forces, which are induced during the seating of the restoration or during functional movements, facilitate pushing of phosphoric acid into the dentinal tubules. In deep preparations, a moderate to severe localized pulpal damage is produced within the first three days, probably because the cement has an initially low pH on setting. The pH of the set cement approximates neutrality only at 48 hours. Resolution of inflammation occurs by 5 to 8 weeks.

The chemical toxicity of zinc phosphate cement materials may not explain the harmful effects on pulp. It is established that bacterial leakage would be an important factor for inducing pulpal damage.

When zinc phosphate is used as a luting agent, an intense diffuse infiltration of neutrophils occurs throughout the pulp tissue (Fig. 31.7). The initial pH of luting mixes ranged from 2.0–3.3, which changes to 3.0–4.2 after one hour. It has been established that such a low pH may induce vascular thrombosis and necrosis in pulp. The pH for zinc phosphate rises during the first 15 minutes. When the cement is used as a base with 1.0 mm remaining dentin, there occurs lifting of the odontoblastic layer and also the inflammatory cells become countable. In case the remaining dentin thickness is 1.5 mm, there occurs only moderate reaction with segmented odontoblastic layer and some reparative dentin formation.

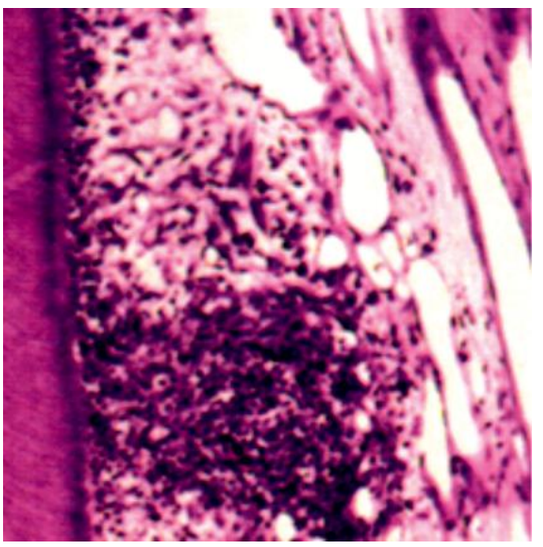


Fig. 31.7: Inflammation adjacent to zinc phosphate cement

C. Composites

The bond strength of composite resin is linked to the etching of the normal mineralized tissue. The effect of etching on pulp is minimum or negligible, since the process is carried out for 10–15 seconds only (Fig. 31.8). To achieve optimal bonding to dentin, the adhesive material (bonding agents) must penetrate the demineralized dentin, enter the dentinal tubules and their branches (Fig. 31.9A, B). The resin monomer enters the collagen fibers to completely infiltrate the demineralized dentin, forming the hybrid layer (Fig. 31.10). Over this, the composite is filled.

The bonding materials have been tried as 'pulp capping agents' with almost satisfactory results. It imparts healing process; however, the polymerization shrinkage may create vacuum in between the remaining dentin and the restoration, subsequently affecting the pulp. It has been established that even after curing, the monomer is leached from composites. This leached monomer is managed by the remaining dentin thickness. In case the dentin thickness is less, the leaching monomer can affect the underlying pulp.

D. Glass-Ionomer Cements

The glass-ionomer cement along with its modified forms is considered biocompatible; mainly indicated in cavity bases and liners. The bonding of glass-ionomer material to dentin involves chemical and mechanical bonds. The chemical bond is based on exchange of ions between carboxyl group of substrate and calcium ions. The mechanical interlocking is based on the demineralization of exposed dentin by polyacrylic acid.

After one week of placement of glass-ionomer cement, the odontoblastic layer is disrupted, along with blood vessels in pulp area (Fig. 31.11). Bacterial

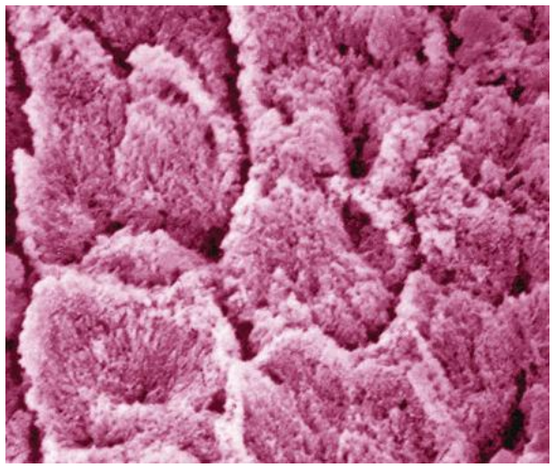


Fig. 31.8: Etched enamel surface. Demineralized rods and inter-rod tissues (×5000)

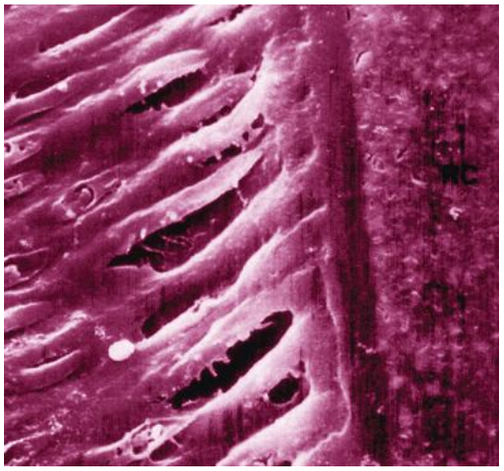


Fig. 31.9A: Dentin hybrid layer and resin tags (×1500)

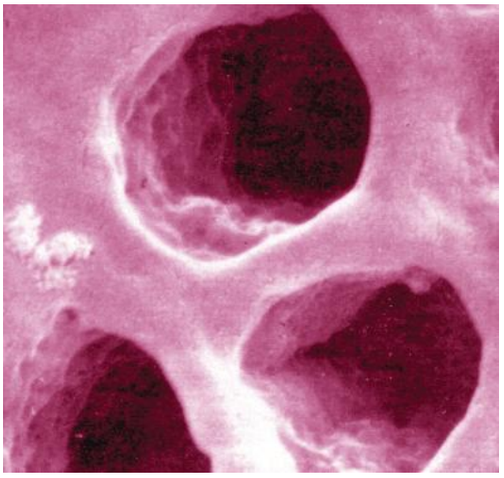


Fig. 31.9B: Dentin etching (open tubules) (×10,000)

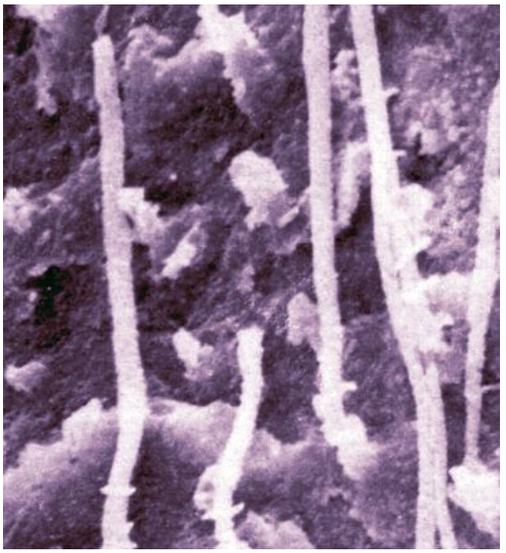


Fig. 31.10: Resin tags and modified smear layer

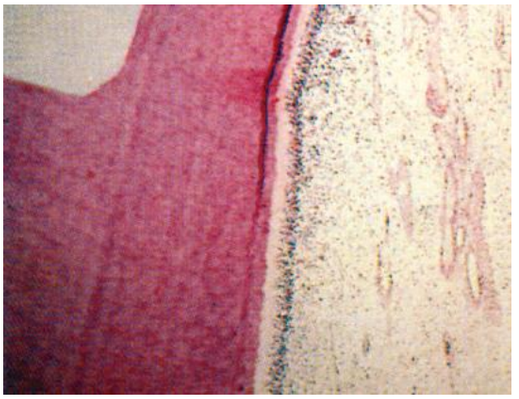


Fig. 31.11: Pulpal reaction with glass-ionomer cement

penetration into tubules is also observed. After about a month, the pulp tissue recovers and displays a normal appearance. The disruption of odontoblasts also become normal.

PULPAL REACTION TO TOOTH PREPARATION

The high-speed cutting instruments utilizing light pressure with intermittent cooling can minimize temperature increase during tooth preparation.

When the cooling is inadequate, the injury leads to displacement of odontoblastic nuclei into the dentinal tubules (Fig. 31.12). The odontoblasts appear disorganized. The burning of dentin becomes evident (Fig. 31.13), if excessive pressure is used and that too without cooling.

The most common feature is the outward movement of contents of the tubules (Fig. 31.14). This movement is dependent on the tubules' openings. Many a times, the smear layer may obturate the opening of the tubules and reduce extrusion of contents of the dentinal fluids. It is established that such flow of fluid is helpful in formation of peritubular dentin. The redistribution of the cellular

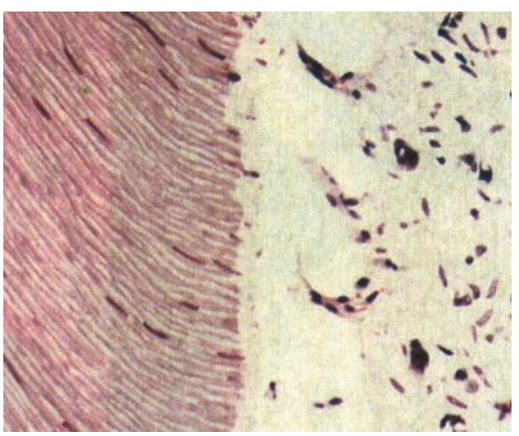


Fig. 31.12: Displacement of odontoblastic nuclei

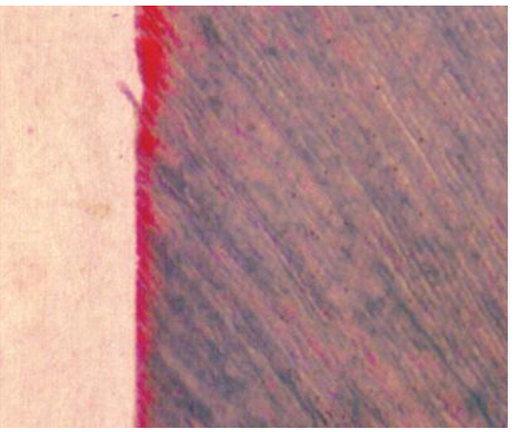


Fig. 31.13: Burning of dentin



Fig. 31.14: Flow of tubular contents

constituents, if continued, leads to degeneration of odontoblastic processes. This might be the basis for formation of 'dead tracts' and subsequently, the tertiary dentin.

Gentle grinding over the dentin leads to increased blood flow; however, grinding in the inner half may lead to decrease in blood flow. The blood flow should decrease after crown preparation, since the inner thickness of dentin might not be more than 1.0 mm (especially in porcelain fused to metal preparation).

In case the cavities are prepared deep and without coolant, vasoconstriction is noted. This vasoconstriction can be due to inhibition of sympathetic nerve stimulation. The flow of dentinal fluid plays an important role. The peripheral flow of fluid following cavity preparation allows plasma proteins to enter the tubules.

The displacement of odontoblastic nuclei into the dentinal tubules has been established. Earlier, 'aspiration of odontoblastic nuclei' was accepted phenomenon, but actually it is the '*displacement of odontoblasts*'. The remaining dentin thickness has always been the key factor in such reactions. A number of morphologic changes occur including intracellular disorganization of the odontoblasts. This can lead to disruption of odontoblastic layer.

The alternative means of cavity preparation, such as air abrasion and laser preparation, also affects the pulp. The generation of heat with the use of lasers is more detrimental.

PULPAL REACTION TO CARIES

The dental caries, may be at any site, affects the underlying pulp. The acidic environment of carious lesions cause both tissue dissolution and in aggressive lesions, cellular injury as well. The initiation of pulpal inflammation also stimulates the repair process. A balance is usually maintained between repair and injury and deviation from the balance leads to changes in the dentin–pulp complex.

Caries create acidic environment by diffusion of bacterial acids. The ability of the cells to withstand this acid environment depends upon hydrogen ion concentration. Local lymphatic drainage may contribute to clearance of acid, but the pulpal lymphatic system is not well developed. The odontoblasts are significantly reduced in size beneath the active enamel lesions; whereas, at the site of arrested lesions, such reduction is not exhibited. As the caries lesion reaches dentinoenamel junction, a significant reduction in cytoplasm to nucleus ratio of odontoblasts is observed. An increase in activity of phosphate enzyme is reported in odontoblasts and in predentin beneath the carious lesions.

The reaction in pulp varies according to intensity and progress of caries; however, the extent of pulpal infection is not systematized.

Pulpal reactions have been reported even with the formation of white-spot lesions. There are marked changes in dentin with 'acute' or rapidly progressing caries than with 'arrested' or slow progressing caries. The odontoblasts in active lesions were significantly smaller than with other lesions. The cellular proliferation of the cell-free zone is also observed in active lesions only. Changes in the subodontoblastic region might include onset of neurogenic inflammatory reactions.

Soon after demineralization, the evidence of tertiary dentin can be noted at the pulp–dentin border, usually defined as '*reactionary dentin*' (Fig. 31.15A). The permeability of dentin plays an important role in these reactions. The initial entry of bacteria, bacterial antigens and toxic components is via dentinal tubules to the pulp.



Fig. 31.15A: Reactionary dentin

The pulp adjacent to deep caries shows the presence of chronic inflammatory exudates, including lymphocytes, macrophages and plasma cells (Fig. 31.15B, C).

The tertiary dentin can be seen at the affected site. The accumulation of inflammatory cells depends upon bacteria associated with caries reaching the tertiary dentin. This stage corresponds to severe pulpal inflammation with minimum chances of healing. It may lead to pulp necrosis at a later stage.

In case of rampant caries, the affected enamel and dentin will break soon, leading to destruction of

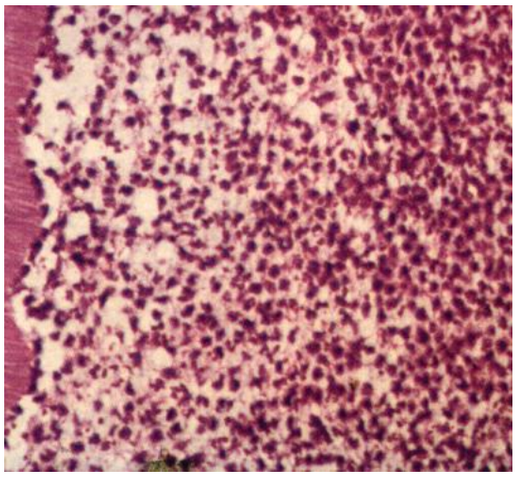


Fig. 31.15B: Polymorphonuclear leukocytes (×100)

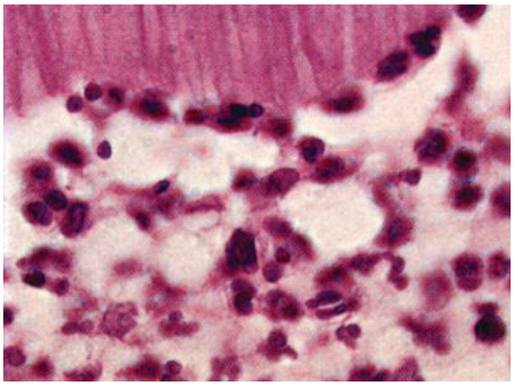


Fig. 31.15C: Leukocytes around dentinal tubules (×250)

odontoblasts and lack of tertiary dentin formation. If the odontoblasts are destroyed slowly, there is formation of hard tissue, which is initially atubular with some cellular inclusions, referred to as '*fibrodentin*' and '*interface dentin*'. The progress of caries, if halted, leads to formation of reparative dentin with differentiation of new secondary odontoblast-like cells. If the lesion is allowed to progress, pulp necrosis may follow.

The immune system helps in defence mechanism of pulp. In association with carious dentin, accumulation of immunocompetent cells has been demonstrated. During rapid lesion progression, accumulation of immunocompetent cells is observed in conjunction with reduced number of primary odontoblasts.

The Flowchart 31.1 illustrates the range of injury, defence and repair events associated with caries.

PULPAL REACTION TO TRAUMA

Physical trauma may lead to fracture of enamel, dentin and exposure of pulp. Orthodontic tooth movements can be performed without any pulpal involvement, but quick orthodontic movements may lead to changes in the pulp–dentin complex.

The extensive trauma involves avulsion or displacement of the jaw bones. The blood supply to the pulp is severed off. Either the pulp is necrosed with time or characterized by formation of hard tissues in the pulp chamber. These hard tissues are referred to as 'osteodentin', because it has cellular infiltration similar to bone. In case of reimplantation, there occur degeneration of odontoblast and simultaneous loss of the adjacent cell-free zone. This degeneration is accompanied by reduction of the width of predentin. In teeth with open apices, the odontoblasts can survive and produce reactionary dentin. An interface-dentin is formed separating primary dentin and the reparative dentin.

The orthodontic movements may lead to increased blood flow in the pulp. Severe hyperemia has been seen in such teeth.

ORTHODONTIC MOVEMENT AND PULPAL REACTION

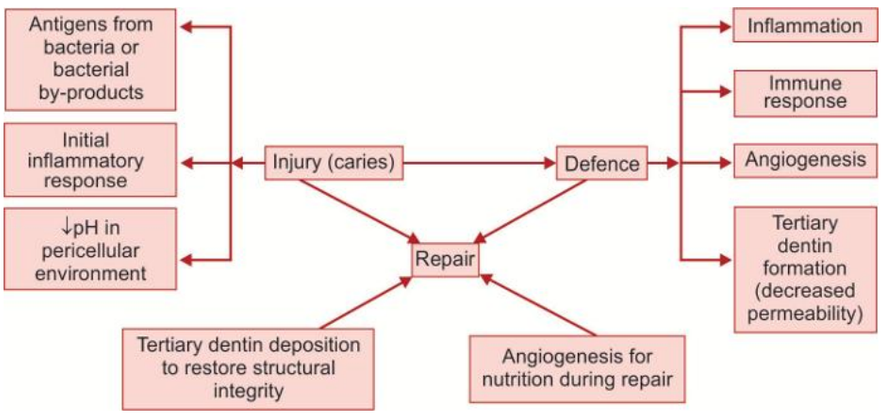
The magnitude of rapidity of orthodontic movement, which may cause pulpal damage, has not been properly documented. The pulpal degeneration may occur after tipping forces or rapid tooth movement, owing to loss of collateral supply to the pulp. The 'tipping force' concept has been refuted, claiming successful rotation of 72°–108° without any pulpal damage. The 'force', however, do affect pulpal vitality. The factors demonstrating the mechanisms, leading to changes in pulp vitality, are as follows:

i. Respiratory rate of the pulpal cells
ii. Blood flow to the pulp
iii. Neural response of the pulp

i. Respiratory Rate of the Pulpal Cells

Respiration is a sign of vitality for any living tissue. Any change in the rate of respiration of any cell affects the state of its vitality. The pulp and its living components are no exception to it. Orthodontic movement depresses the respiratory rate by a mean

Flowchart 31.1: Repair and defence mechanism in caries



value of 27.4%. A few authors have observed a significant correlation between amount of decrease in pulp tissue respiration rate and the age of the patient. This rate of respiratory depression is more in older patients due to the constriction of their apical foramina that decreases the blood flow. This also explains why younger teeth show less effect than the older ones. The changes in tissue respiration (possibly hypoxia), because of orthodontic movement of tooth, cause increase in aspartate aminotransferase (AST) activity level and affect pulp tissue, changing its neural response. It has also been opined that decrease of some protein expressions block the regeneration of pulp structure. Radiorespirometric methods have confirmed respiratory depression in pulp, which is under orthodontic tooth movement.

ii. Blood Flow to the Pulp

It is hypothesized that dynamic mechanisms exist in the control of blood flow in a tooth undergoing orthodontic movement. When an orthodontic force is applied, pulp tissue reacts with initial pulp hyperemia, subsequently proceeding to degranulation of mast cells characterized by cell damage and allied biochemical reactions. This is a classical feature of acute inflammation (Fig. 31.16). The initial stage of reactive hyperemia usually reverts back to normal in due course of time. When the pulp of teeth subjected to orthodontic forces was grown in culture, it showed sprouting of new vessels. Angiogenic changes have been observed after orthodontic movement (increase in number of microvessels), expressing rise of angiogenic growth factor in pulp. The factors responsible for such changes are platelet-derived growth factor, vascular endothelial growth factor and transforming growth factor-β. Transforming growth factors (TGF-β1 and TGF-β3) are responsible for stimulation of reactionary dentin and also reparative dentin after orthodontic tooth movement. Since dental pulp is limited in narrow area, the increase blood flow (dilatation of arterioles) causes increase in pulp pressure and compression of venous flow. It is established that most of the changes in pulpal blood are reversible (during tooth movement); only if pulp is not irritated earlier because of deep caries, trauma, etc. It is also asserted that fully developed (mature apex) teeth are sensitive to irreversible pulp inflammation and immature apex teeth are not sensitive. It is established that application of severe force for a longer period may affect pulpal blood flow than short-term application of same force. The two variables of decrease in pulpal respiration and an altered blood flow interplay to bring about dystrophic mineralization with deposition of reparative dentin in coronal and radicular portions of the tooth. It is established that such changes beginning at the peri apex influence the odontoblasts in the formed teeth and Hertwig's epithelial root sheath (HERS) in the developing teeth (Fig. 31.17). Such changes, especially when accompanied with a history of previous trauma, contribute to the mechanism of loss of pulp vitality after orthodontic movement.

iii. Neural Response of the Pulp

Neural response during orthodontic tooth movement has also been evaluated. A few authors examined pulp axon response to orthodontic tooth movement and observed no significant change in teeth treated with conservative orthodontic treatment. Calcitonin generated peptide (CGP), a neuropeptide released from C-type nerve fibers of pulp after injury, has the ability to initiate vasodilatation along with regulating inflammatory cells (macrophages, mast cells and lymphocytes). CGP has been shown to increase in

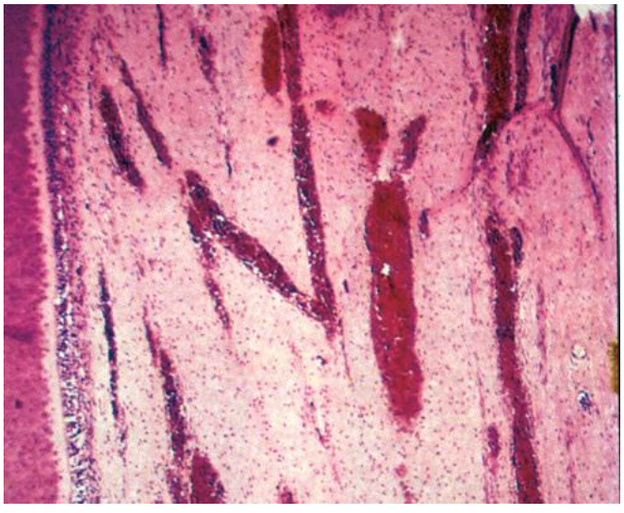


Fig. 31.16: Reactive hyperemia

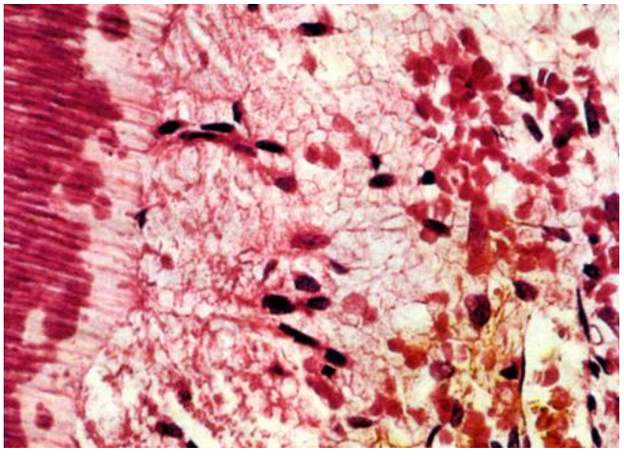


Fig. 31.17: Hemorrhage in the odontoblastic layer

teeth under orthodontic force as compared to normal neuropeptide values. The decrease in the neural mediators causes a decreased sensation of pain. The partial to complete diminution of the pulp space may be evident on radiographs. A distinguishing feature is that such obliterations affect the apical third first followed by coronal aspects of the canal. Any case presenting with altered sensation must be considered a manifestation of post-orthodontic pulpal damage. The healing in such cases is believed to be normal. Orthodontic movement may not hinder periapical healing, but it certainly delays the process.

PULPAL REACTION TO VITAL BLEACHING

The vital tooth bleaching incorporates carbamide solution in different modifications. Initial reaction of the pulp to bleaching includes less distinction of cell-free zone. In areas on facial surface of these teeth, the cells migrate to these areas. Scattered leukocytes are observed along with irregularities in the pseudostratified odontoblastic layer. Only slight pulp reaction is observed (Fig. 31.18A, B). In any case even with high concentration, carbamide does not produce moderate or severe reactions. A few authors have opined that higher concentration of hydrogen peroxide and heat is potentially harmful to pulp. The changes in pulp tissue are reversible within two weeks. Therefore, two weeks of treatment with 10% carbamide peroxide used for nightguard vital bleaching is considered safe for the pulp.

A 3.0% solution of hydrogen peroxide is capable of causing transient reduction in pulpal blood circulation and occlusion of pulpal blood vessels. The most common side effect experienced by patients using home bleaching technique is mild, transient temperature sensitivity. This sensitivity appears to be dose related rather than pH related.

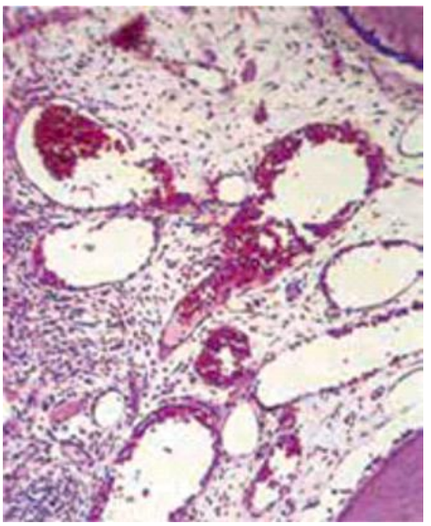


Fig. 31.18A: Hyperemic pulp after bleaching

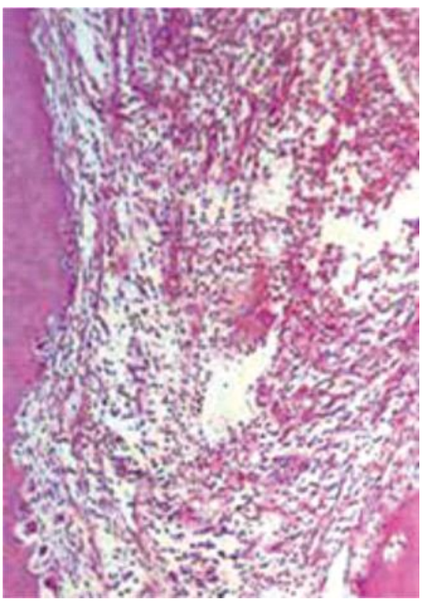


Fig. 31.18B: Inflammation after bleaching

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